PERFECT MODULES ARE EXACTLY THE MODULES WITH A FINITE PROJECTIVE RESOLUTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted statement span and the counted proof span each contain a displayed resolution; both are delivered verbatim, including the author's slight notational shift from F_i in the statement to E^{-i} in the proof. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The characterization of perfect complexes and the pseudo-coherence of modules. invoked at proof lines 18932-18935. Produces the finite free resolution E^bullet of M.
- The last-term-flat criterion bounding flatness of a cokernel in a resolution. invoked at proof lines 18936-18938. Yields the index d at which F_d becomes flat.
- A finitely presented flat module is finite projective. invoked at proof lines 18938-18940. Turns flatness of F_d into finite projectivity, giving the finite resolution.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 16883: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16883–16883. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 16003: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16003–16003. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 18889: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18889–18889. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let M be a module over a ring R. The following are equivalent*

(1) *M is a perfect module, and*
(2) *there exists a resolution*

$$0 \to F_d \to \ldots \to F_1 \to F_0 \to M \to 0$$

with each F_i a finite projective R-module.

Proof. Assume (2). Then the complex $E^\bullet$ with $E^{-i} = F_i$ is quasi-isomorphic to $M[0]$. Hence M is perfect. Conversely, assume (1). By Lemmas 18889 and 16003 we can find resolution $E^\bullet \to M$ with E^{-i} a finite free R-module. By Lemma 16883 we see that $F_d = \mathrm{Coker}(E^{d-1} \to E^d)$ is flat for some d sufficiently large. By Algebra, Lemma 00NX we see that F_d is finite projective. Hence

$$0 \to F_d \to E^{-d+1} \to \ldots \to E^0 \to M \to 0$$

is the desired resolution. □